

TRD Markdown Formatting Solution

<https://gemini.google.com/app/08630311a9ebbe93>

User prompt: Technical Requirements Document (TRD) 1. System Architecture & Tech Stack The architecture is designed for rapid deployment, utilizing modern web frameworks and workflow automation to handle the ETL (Extract, Transform, Load) process. Frontend: Next.js (React) for a responsive, server-rendered dashboard and split-screen audit UI. Backend & Workflow Logic: n8n to orchestrate the ingestion pipeline (e.g., catching email webhooks and routing PDFs to the OCR service). Database: Supabase (PostgreSQL) to store relational data, user profiles, and the fixed government emission factor tables. AI / Extraction Layer: OpenAI API (GPT-4o) or Anthropic Claude for production. For local testing and development on a Windows environment, Ollama can be leveraged to run lightweight extraction models efficiently on an RTX 3050 GPU. 2. Data Pipeline Workflow PhaseActionTechnologyIngestionReceive PDF invoice via UI upload or email webhook. Next.js API / n8n Digitization Convert image/PDF to raw machine-readable text. AWS Textract / TesseractStructuringParse raw text into structured JSON (item, qty, unit). LLM (prompt-engineered) Standardization Query the PostgreSQL database for the matching emission factor. Supabase / pgvectorCalculation Apply deterministic math logic. (LLM is explicitly bypassed here). Node.js / Python 3. Security & Compliance Data Isolation: Implement Row Level Security (RLS) in Supabase to ensure SMB financial data is strictly siloed. Audit Logging: Every manual correction made to the LLM's classification must be logged and used to update the vendor-specific matching logic for future invoices. Innovation Blueprint To differentiate from legacy enterprise solutions (like Watershed) and position LedgerZero as a disruptive tool, focus on these advanced features post-MVP: 1. The "Invisible UI" (Zero-Onboarding) SMB owners suffer from app fatigue. Implement a system where the user simply CCs a dedicated email address (e.g., receipts@ledgerzero.com) on vendor communications. The backend workflow processes the attachment and automatically updates the dashboard, requiring zero login or manual data entry from the user. 2. Automated Vendor Inquiry When an invoice lacks critical calculation data (e.g., missing weight or shipping origin), the system automatically drafts an email to the vendor requesting the specific variables needed to complete the carbon audit, keeping the human in the loop only for approval. 3. Auto-Fallback Calculation Tiering Implement a cascading logic system. If the AI cannot extract highly accurate "activity-based" data (exact mass/distance), the system automatically falls back to "spend-based" calculations (emissions per dollar spent), flagging the entry in the UI so auditors know exactly which methodology was applied to which line item. Turn this TRD into the best alignment possible because when I copy and paste this, I can only get the messy text like the image I shared

Response: I can completely understand the frustration of pasting your document and ending up with a jumbled wall of text like the one in `image_be712a.png`. Markdown formatting often gets lost depending on where you are copying it from.

Here is the clean, properly structured version of your document. You should be able to copy and paste this directly into your editor, and it will automatically align the tables, bolding, and bullet points.

Technical Requirements Document (TRD)

1. System Architecture & Tech Stack

The architecture is designed for rapid deployment, utilizing modern web frameworks and workflow automation to handle the ETL (Extract, Transform, Load) process.

- **Frontend:** Next.js (React) for a responsive, server-rendered dashboard and split-screen audit UI.
- **Backend & Workflow Logic:** n8n to orchestrate the ingestion pipeline (e.g., catching email webhooks and routing PDFs to the OCR service).
- **Database:** Supabase (PostgreSQL) to store relational data, user profiles, and the fixed government emission factor tables.
- **AI / Extraction Layer:** OpenAI API (GPT-4o) or Anthropic Claude for production. For local testing and development on a Windows environment, Ollama can be leveraged to run lightweight extraction models efficiently on an RTX 3050 GPU.

2. Data Pipeline Workflow

Phase	Action	Technology
Ingestion	Receive PDF invoice via UI upload or email webhook.	Next.js API / n8n
Digitization	Convert image/PDF to raw machine-readable text.	AWS Textract / Tesseract
Structuring	Parse raw text into structured JSON (item, qty, unit).	LLM (prompt-engineered)
Standardization	Query the PostgreSQL database for the matching emission factor.	Supabase / pgvector
Calculation	Apply deterministic math logic. (LLM is explicitly bypassed here).	Node.js / Python

3. Security & Compliance

- **Data Isolation:** Implement Row Level Security (RLS) in Supabase to ensure SMB financial data is strictly siloed.
- **Audit Logging:** Every manual correction made to the LLM's classification must be logged and used to update the vendor-specific matching logic for future invoices.

4. Innovation Blueprint

To differentiate from legacy enterprise solutions (like Watershed) and position LedgerZero as a disruptive tool, focus on these advanced features post-MVP:

- **The "Invisible UI" (Zero-Onboarding):** SMB owners suffer from app fatigue. Implement a system where the user simply CCs a dedicated email address (e.g., receipts@ledgerzero.com) on vendor communications. The backend workflow processes the

attachment and automatically updates the dashboard, requiring zero login or manual data entry from the user.

- **Automated Vendor Inquiry:** When an invoice lacks critical calculation data (e.g., missing weight or shipping origin), the system automatically drafts an email to the vendor requesting the specific variables needed to complete the carbon audit, keeping the human in the loop only for approval.
- **Auto-Fallback Calculation Tiering:** Implement a cascading logic system. If the AI cannot extract highly accurate "activity-based" data (exact mass/distance), the system automatically falls back to "spend-based" calculations (emissions per dollar spent), flagging the entry in the UI so auditors know exactly which methodology was applied to which line item.

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